EXP8:WIRESHARK SEQ ANALYSIS

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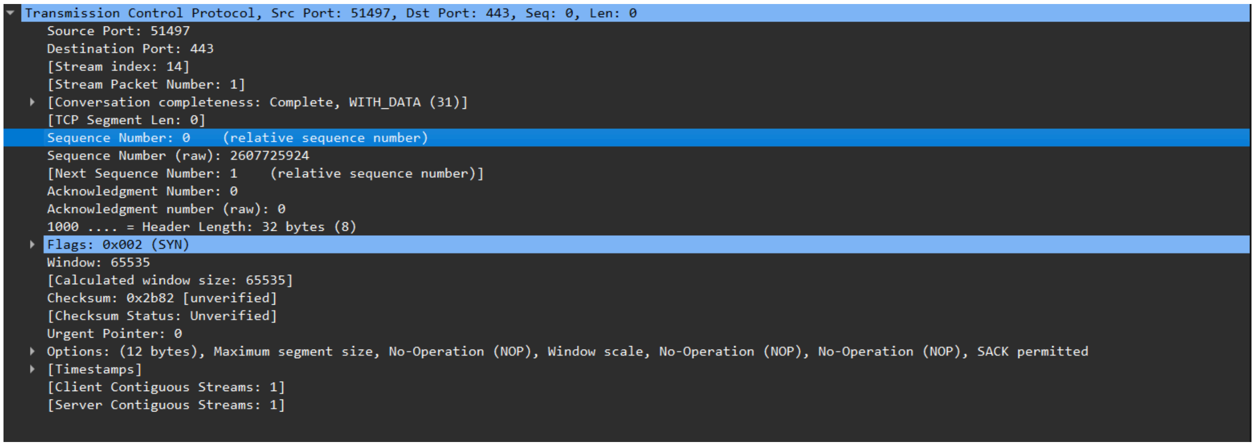
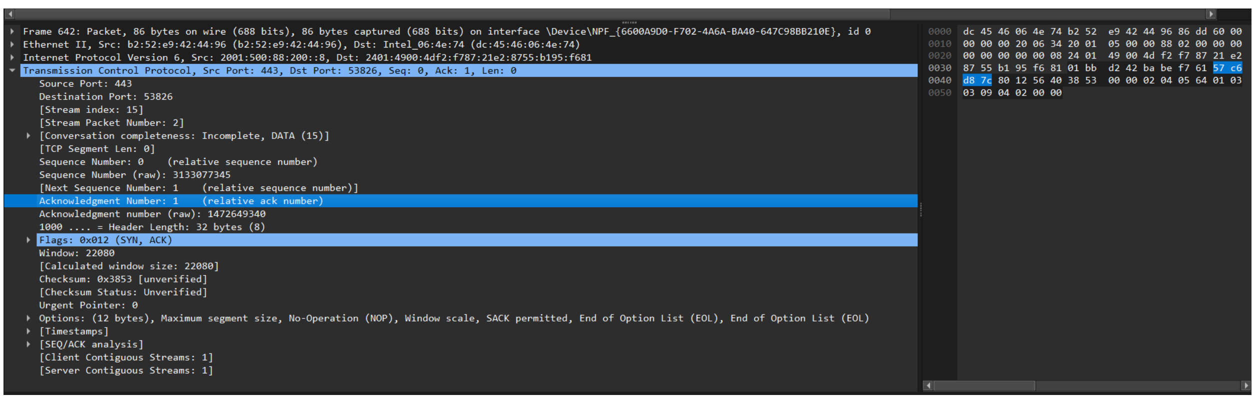
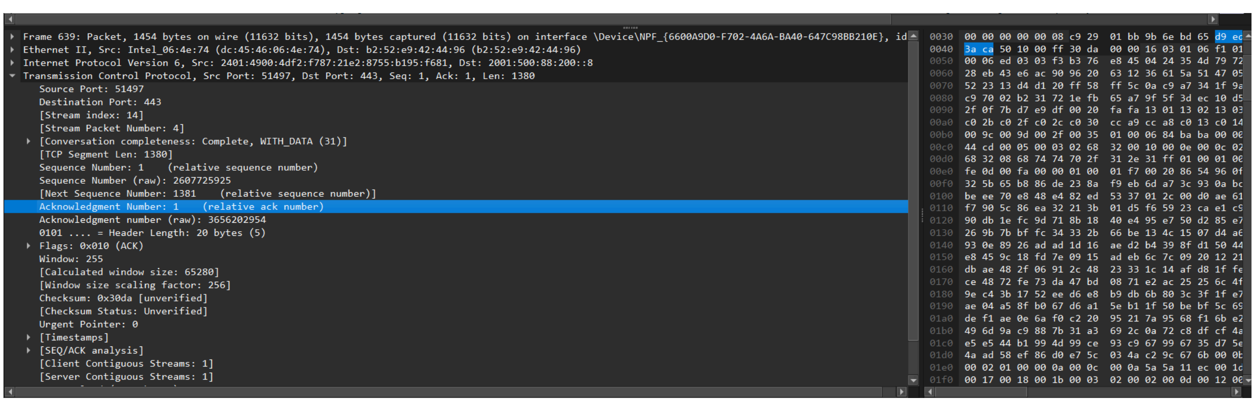
RELATIVE SEQUENCE NUMBER IS ZERO FOR (SYN)



RELATIVE SEQ=0 AND ACK=1(SYN-ACK)



RELATIVE SEQ=1 AND ACK=1(ACK)



DEMONSTRATION OF ATTACKS:

2016	73.014047	10.77.167.47	35.228.141.16	TCP	66			60533 → 443 [SYN] Seq=0
2017	73.225803	35.228.141.16	10.77.167.47	TCP	70	2016	0.211756000	443 → 60533 [SYN, ACK]
2018	73.225943	10.77.167.47	35.228.141.16	TCP	54	2017	0.000140000	60533 → 443 [ACK] Seq=1
2019	73.226574	10.77.167.47	35.228.141.16	TCP	1454			60533 → 443 [ACK] Seq=1
2020	73.226574	10.77.167.47	35.228.141.16	TLSv1…	446			Client Hello (SNI=e2c13
2021	73.233325	2603:1040:a06:3:…	2401:4900:4df2:f…	TLSv1…	105			Application Data

THE [SYN,ACK] GREY COLOR SHOWS THE You will see many "SYN" packets without any corresponding "SYN-ACK" or "ACK."

2117	78.510357	2001:4860:4827:7…	2401:4900:4df2:f…	UDP	92			443 → 54266 Len=26
2118	78.691025	35.228.141.16	10.77.167.47	TLSv1…	97			Application Data
2119	78.691306	10.77.167.47	35.228.141.16	TCP	54	2118	0.000281000	60533 → 443 [FIN, ACK] Seq=2559 Ack=4959 Win=65280 Len=
2120	78.911743	35.228.141.16	10.77.167.47	TLSv1…	82	2119	0.220437000	Application Data
2121	78.911743	35.228.141.16	10.77.167.47	TCP	58			443 → 60533 [FIN, ACK] Seq=4983 Ack=2560 Win=61952 Len=
2122	78.911855	10.77.167.47	35.228.141.16	TCP	54	2120	0.000112000	60533 → 443 [RST, ACK] Seq=2560 Ack=4983 Win=0 Len=0
2123	79.620102	10.77.167.47	10.77.167.180	DNS	77			Standard query 0x9b57 AAAA beacons3.gvt2.com
2124	79.638987	10.77.167.180	10.77.167.47	DNS	77			Standard query response 0x9b57 AAAA beacons3.gvt2.com
2125	80.172171	2401:4900:4df2:f…	2404:6800:4007:8…	QUIC	801			Protected Payload (KP0), DCID=fd0c7efed1d74aa6

Packets with the [RST] flag set in red.Sends Reset (R) flags to force a connection to close.

3035	140.358816	2401:4900:4df2:f…	2404:6800:4007:8…	UDP	91			53214 → 443 Len=29
3036	140.459011	2404:6800:4007:8…	2401:4900:4df2:f…	UDP	87			443 → 53214 Len=25
3037	140.940239	2603:1061:f:100:…	2401:4900:4df2:f…	TCP	78			443 → 49321 [RST, ACK] Seq=5745 Ack=3393 Win=0 Len=0
3038	140.980731	2401:4900:4df2:f…	2603:1040:a06:3:…	TLSv1…	105			Application Data
3039	141.062104	2603:1040:a06:3:…	2401:4900:4df2:f…	TCP	78	3038	0.081373000	443 → 51780 [ACK] Seq=244 Ack=451 Win=536 Len=0
3040	141.087355	2603:1061:e::254	2401:4900:4df2:f…	TCP	78			443 → 50895 [RST, ACK] Seq=5747 Ack=4295 Win=0 Len=0
3041	141.172701	2401:4900:4df2:f…	2001:4860:4827:7…	UDP	859			54266 → 443 Len=797
3042	141.262949	2001:4860:4827:7…	2401:4900:4df2:f…	UDP	99			443 → 54266 Len=33
3043	141.275350	2401:4900:4df2:f…	2001:4860:4827:7…	UDP	95			54266 → 443 Len=33
3044	141.306895	2001:4860:4827:7…	2401:4900:4df2:f…	UDP	757			443 → 54266 Len=691
3045	141.308664	2001:4860:4827:7…	2401:4900:4df2:f…	UDP	110			443 → 54266 Len=44
3046	141.321481	2401:4900:4df2:f…	2001:4860:4827:7…	UDP	102			54266 → 443 Len=40
3047	141.364228	34.1.15.89	10.77.167.47	TCP	58			[TCP Keep-Alive] 443 → 55795 [ACK] Seq=4805 Ack=1889 Wi
3048	141.364329	10.77.167.47	34.1.15.89	TCP	54			[TCP Keep-Alive ACK] 55795 → 443 [ACK] Seq=1889 Ack=480
3049	141.382703	2001:4860:4827:7…	2401:4900:4df2:f…	UDP	92			443 → 54266 Len=26
3050	143.094354	2401:4900:4df2:f…	2603:1046:c04:89…	TCP	74			65376 → 443 [RST, ACK] Seq=2391 Ack=6377 Win=0 Len=0
3051	143.212967	2404:6800:4007:8…	2401:4900:4df2:f…	UDP	103			443 → 65022 Len=37
3052	143.234058	2401:4900:4df2:f…	2404:6800:4007:8…	UDP	95			65022 → 443 Len=33
3053	145.195373	2404:6800:4007:8…	2401:4900:4df2:f…	UDP	219			443 → 53214 Len=157
3054	145.218582	2401:4900:4df2:f…	2404:6800:4007:8…	UDP	95			53214 → 443 Len=33
3055	145.273659	2404:6800:4007:8…	2401:4900:4df2:f…	UDP	219			443 → 53214 Len=157
3056	145.274199	2401:4900:4df2:f…	2404:6800:4007:8…	UDP	96			53214 → 443 Len=34
3057	147.203753	10.77.167.47	103.10.124.5	TLSv1…	108			Application Data
3058	147.326727	103.10.124.5	10.77.167.47	TCP	54	3057	0.122974000	443 → 63919 [ACK] Seq=2497 Ack=498 Win=1869 Len=0
3059	148.268310	204.79.197.222	10.77.167.47	TCP	58			443 → 49667 [RST, ACK] Seq=5191 Ack=4656 Win=0 Len=0
3060	151.134360	2401:4900:4df2:f…	2603:1063:14::128	TLSv1…	124			Application Data
3061	151.288559	2603:1040:a06:3:…	2401:4900:4df2:f…	TLSv1…	105			Application Data
3062	151.319283	2603:1063:14::128	2401:4900:4df2:f…	TLSv1…	113	3060	0.184923000	Application Data

You send a SYN to a "Reflector" server, but you spoof the source address as the "Victim." The Reflector sends the SYN-ACK to the Victim, not you.Incoming SYN-ACKs to a machine that never sent a SYN.